

Travellers' Acceptance of Consumer-Generated Media: An Integrated Model of Technology Acceptance and Source Credibility Theories

Julian K. Ayeh*

Abstract

Despite the rising influence of social media, the use of consumer-generated media (CGM) for the task of travel planning still meets with scepticism among certain online travel consumers. Hence the need to further explore the psychological factors underlying this aspect of online behaviour. The study proposes a model of consumer-generated media acceptance for the purpose of travel planning which integrates the Technology Acceptance Model with the Source Credibility Theory. Using an online survey of 661 valid responses and structural equation modelling, the findings highlight the critical factors relevant to the cognitive processes which determine online travellers' affective and conative responses to the use of consumer-generated media for travel planning. The results suggest that integrating technology acceptance factors with that of information adoption can enhance the understanding of consumer-generated media usage in the vacation planning context. The study outcome holds implications for theory and practice.

Keywords: User-Generated Content, Social Media, Source Credibility, Structural Equation Modelling, TripAdvisor.

1 Introduction

The rapidly increasing popularity of social media for the exchange of information cannot be ignored. More and more people are employing social media tools to gather information, share experiences and discuss concerns (Litvin, Goldsmith, & Pan, 2008; Burgess, Sellitto, Cox & Buultjens 2009; Yoo & Gretzel, 2011). A momentous portion of these concerns relates to travel planning, as the travel planning process is frequently regarded as one of uncertainty (Cox, Burgess, Sellitto, & Buultjens, 2009; Xiang, & Gretzel, 2010). This essentially arises from the very nature of travel products: characteristics of tourism as a service product – intangibility, perishability, heterogeneity, and simultaneous production and consumption – often result in higher perceived risks with barely any objectively measurable qualities prior to purchase (Hsu, Kang & Lam, 2006). For this reason, prospective travellers are inclined to rely on word-of-mouth communications from friends and relatives (who have previously experienced the travel product) in making their travel-related choices (Litvin et al., 2008; Burgess et al., 2009).

In the 21st century, however, the emergence of Web 2.0 and the growing interest in social media have led to the transfer of this “word-of-mouth” phenomenon to online domains. Travellers are increasingly resorting to various social media platforms for consumer-generated media (CGM) to inform their travel-related decisions (Yoo & Gretzel, 2011). Consequently, CGM websites have emerged as one of the most prominent platforms for improving travel information accessibility and for enhancing travel decision making (Litvin et al., 2008). Yet regardless of the growing interest in CGM and social media in general, the use of CGM for travel planning still meets with scepticism from some

travellers (Dellarocas, 2003; Burgess *et al.*, 2009; Cox *et al.*, 2009). Thus, it is important to understand the conditions under which CGM will be embraced and utilised by online travel consumers. While social media has recently received considerable attention in both academia and industry, research on CGM has predominantly focused on its implications for product awareness, purchase decisions, sales and complaint management (e.g. Arsal, Backman, & Baldwin, 2008; Yoo & Gretzel, 2008; Au, Buhalis & Law, 2009; Cox *et al.*, 2009; Gupta & Harris, 2010; Huang, Chou, & Lin, 2010; Ye, Law, Gu, & Chen, 2011) with less attention on the determinants of CGM usage for specific tasks like travel planning. The present study therefore seeks to shed light on the psychological and decision making mechanisms underlying the use of CGM for travel planning.

Traditionally, information systems acceptance has been examined with various theories (e.g. innovation diffusion theory, determinants of ICT assimilation model, motivation theory, theory of planned behaviour, etc.). Information Systems literature demonstrates that Davis' (1986) Technology Acceptance Model (TAM) is the most prevalent and compares favourably with alternative models (see Lin (2007) and Venkatesh (1999) for review). However, given the discrete nature of CGM as well as the high involving task of travel planning, the TAM might not offer sufficient understanding of users' attitudes and intentions. Therefore, in the context of CGM adoption for travel planning, which factors can best explain acceptance? The study attempts to address this research question by proposing and empirically testing an integrated model which draws on previous literature that identifies the antecedents of technology acceptance as well as information adoption to predict attitudes and intentions towards using CGM for travel planning.

Given that the particular context of CGM raises credibility concerns such as the subjective nature of online consumer opinions (Dellarocas, 2003), the complicated task of judging the views of complete strangers (Litvin *et al.*, 2008; Burgess *et al.*, 2009), vulnerability to strategic manipulation and abuse (Dellarocas, 2003; Litvin *et al.*, 2008) and uncertainty about the trustworthiness of the website operators (Gretzel, 2006), this study proposes that, integrating the Source Credibility Model (Hovland, Janis & Kelley, 1953) with the classical TAM offers a more salient understanding of online travellers' acceptance of CGM for travel planning. By utilising technology acceptance literature to integrate and add to relevant concepts from the Source Credibility Theory, this study is expected to offer a deeper understanding of online travellers' acceptance of CGM.

2 Theory and Hypotheses

Several studies have sought to understand online travellers' behaviour regarding CGM and social media in general. Early attempts focused on online community participation (e.g. Wang & Fesenmaier, 2004a, 2004b). Some studies have addressed the implications of CGM for travel decisions (e.g. Aarsal *et al.*, 2008; Yoo & Gretzel, 2008; Cox *et al.*, 2009). Recent studies have widened further to address the various forms of user-generated content (e.g. Enoch & Grossman, 2010; Huang, Chou, & Lin, 2010; Chen, Shang, & Li, 2014). While these studies clarify the understanding of online travellers' behaviour regarding social media platforms within the online travel domain, the determinants of travellers' acceptance of CGM for travel planning still deserve further attention. Building on previous research, the present study investigates how the integration of theories from two distinct

disciplines can help to shed light on the factors underlying online travellers' acceptance of CGM for travel planning. The CGM platform can be construed as a travel information system (involving a technological application as well as a source of information). Hence factors which determine technology acceptance and those which influence information acceptance may both be of relevance to the understanding of this phenomenon. The current study therefore proposes a model which integrates Davis' (1986) TAM with Hovland et al.'s (1953) Source Credibility Model to explain online travellers' intentions to use CGM for travel planning.

2.1 *Technology Acceptance Theory*

In the existing literature, several theoretical perspectives have been applied in an attempt to understand individuals' acceptance and use of technology-related applications. Of these, the TAM stands out as the most parsimonious and, perhaps, the most influential approach to studying user acceptance. The model was originally proposed by Davis (1986) based on the Theory of Reasoned Action (Ajzen & Fishbein, 1980). The theory postulates that individual's acceptance of information systems are determined by two cognitive factors (i.e. perceptions of usefulness and ease of use). TAM has been widely applied to investigate user-acceptance of various types of technology including smart phones (e.g. Joo & Sang, 2013), technology based services (e.g. Zhu & Chan, 2014), e-learning (e.g. Persico, Manca, & Pozzi, 2014) and the new media (e.g. Workman, 2014). In the locale of hospitality and tourism-related research, TAM applications include employees' acceptance of IT in upscale hotels (Lam, Cho, & Qu, 2007), hotel front office systems (Kim, Lee, & Law, 2008) and other information systems (Huh, Kim, & Law, 2009), restaurant computing

The findings of these studies demonstrate that perceived usefulness and ease of use are important determinants of user acceptance of technology-related applications. Davis (1989) defines perceived usefulness as “the degree to which a person believes that using a particular system would enhance his or her job performance” (p. 320). In the present study, perceived usefulness can be construed as the extent to which an online travel consumer believes that using CGM enhances his or her travel planning. The existing literature suggests that perceived usefulness drives attitude and usage intention (e.g. Ayeh, Au & Law, 2013; Joo & Sang, 2013; Persico et al., 2014; Workman, 2014). Prospective travel consumers who perceive CGM to be useful are more likely to have favourable affective and conative responses towards its adoption for travel planning than those who do not. Hence the hypotheses,

Hypothesis 1. Perceived usefulness has a significant positive effect on attitude.

Hypothesis 2. Perceived usefulness has a significant positive effect on usage intention.

Perceived ease of use has been conceptualised by Davis (1989) as “the degree to which a person believes that using a particular system would be free of effort” (1989, p. 320). In the present study, ease of use refers to the extent to which the online travel consumer believes that using CGM for travel planning will be free from effort. Prior research suggests that ease of use influences attitude (Zhu & Chang, 2014). Also, it is generally believed that a system will be regarded as more useful if it is easy to use. For example, studies by

Morosan (2012) and Kim *et al.* (2008) among others, offer strong empirical support for a positive relationship between perceived usefulness and ease of use. The present study therefore proposes that

Hypothesis 3. Perceived ease of use has a significant positive effect on attitude towards usage.

Hypothesis 4. Perceived ease of use has a significant positive effect on perceived usefulness.

2.2 Source Credibility Theory

The proposed model also considers the Source Credibility Theory (Hovland *et al.*, 1953). While the source credibility perspective is expected to be relevant in the study of individuals' acceptance and use of information sources, it can be expected to be particularly useful in the context of CGM because of the many concerns regarding CGM which largely hinge on credibility (e.g. Dellarocas, 2003; Litvin *et al.*, 2008).

Source credibility has received extensive attention in the Marketing and Communication Studies literatures as regards the study of various phenomena (for review, see Metzger, Flanagin, Eyal, Lemus, & McCann, 2003; Pornpitakpan, 2004). In the marketing literature, source credibility theory has often been used to examine the effectiveness of celebrity endorsement. Researchers in Communication Studies also tend to apply this theory to compare the credibility of different media channels (Johnson & Kaye, 2009). In the online domain, more recently, researchers have studied source credibility in the context of different theories to understand its role in online behaviour. For example, Sussman and

Siegal (2003) examined source credibility within the framework of the Elaboration Likelihood Model (ELM; Petty & Cacioppo, 1986) and found source credibility to be significantly related to information usefulness in the context of knowledge transfer via e-mails. Similarly, Zhang and Watts (2008) employed the source credibility perspective in examining information adoption in online communities, whereas Li (2013) investigated the influence of source credibility on information system acceptance.

Most studies on source credibility have employed expertise (or competence) and trustworthiness as key dimensions (Fogg & Tseng, 1999). These two factors also appear to be the most relevant to the particular context of CGM. Ohanian (1991) defines trustworthiness as the “consumer's confidence in the source for providing information in an objective and honest manner” (p. 47). In the present study, the source here refers to travel consumers who post CGM. Perceived expertise, on the other hand, describes the extent to which CGM contributors are perceived to be “a source of valid assertions” (Hovland *et al.*, 1953, p. 21). Perceived expertise, in this study's setting, concerns online travellers' perceptions of the CGM contributors' prior experiences with the travel product and their ability to make meaningful evaluations. Early research underscores the importance of source credibility in determining responses towards information sources (Hovland *et al.*, 1953). More recent studies in various contexts have also accentuated the significant effects of source trustworthiness on various consumer outcomes such as attitudes toward a message, information adoption and purchase intentions (e.g., MacKenzie & Lutz, 1989; Sussman & Siegal, 2003; Pornpitakpan, 2004; Jin, Cheung, Lee, & Chen, 2009; Wang & Doong, 2010; Li, 2013; Kim & Kim, 2013; Ayeh, Au, & Law, 2013b). Thus, it is expected that the extent of trustworthiness allotted to travel consumers who post

CGM will influence online travellers' perception of CGM usefulness, as well as their attitudes and usage intention in the context of travel planning. Hence, the hypotheses:

Hypothesis 5. Perceived trustworthiness has a significant positive effect on perceived usefulness.

Hypothesis 6. Perceived trustworthiness has a significant positive effect on attitude towards usage.

Hypothesis 7. Perceived trustworthiness has a significant positive effect on usage intention.

Extant research underscores the role of expertise in determining attitudes and intentions towards information sources (Pornpitakpan, 2004). A study carried out by Ohanian (1991) demonstrates that perceived expertise positively affects attitude change in the context of celebrity endorsement advertising. In online contexts, Jin *et al.* (2009) and Sussman and Siegal (2003) have also established a positive relationship between source expertise as a credibility dimension and information usefulness. Given the Source Credibility Theory (Hovland et al., 1953), perceived expertise is thus expected to influence online travel consumers' attitudes toward the use of CGM for travel planning as well as their CGM usefulness perception. The following hypotheses are therefore posited:

Hypothesis 8. Perceived expertise has a significant positive effect on perceived usefulness.

Hypothesis 9. Perceived expertise has a significant positive effect on attitude towards usage.

Hypothesis 10. Perceived expertise has a significant positive effect on usage intention.

Attitude is generally held as a prime predictor of usage intention. The TAM, the TPB and many social psychology theories recognise the salience of attitude in determining behavioural intention. In online settings, the influence of attitude on intention has also been found to be valid (e.g. Bruner & Kumar, 2000; Castañeda *et al.*, 2009). Therefore, the present study hypothesises that:

Hypothesis 11. Attitude towards usage has a significant positive effect on usage intention.

Behavioural intention is the final endogenous construct in the proposed model. In the current study, it represents online travellers' conative responses to the use of CGM for travel planning. Behavioural intention has generally been regarded as "a person's subjective probability that he will perform some behaviour" (Fishbein & Ajzen, 1975, p. 288). A definition by Warshaw and Davis (1985) emphasised the extent of planning related to performing a particular behaviour; "the degree to which a person has formulated conscious plans to perform or not perform some specified future behaviour" (p. 214). The 'specified future behaviour', in this study's context, is the use of CGM for travel planning. Figure 1 depicts the proposed model with the hypothesised relationships.

[Please insert Fig. 1 here: Proposed model with hypothesised relationships]

3 Methods

The unit of analysis in the present study is the individual travel consumer who employs the Internet for travel information search; that is, a potential user of CGM. Data to empirically test the specified hypotheses were collected through an online survey.

3.1 Measurement

Scales were derived from previous literature and, where possible, existing validated scales were adopted. Scales for *perceived usefulness*, *ease use* and *usage intention* were modified from existing TAM research (Davis, 1989; Davis, Bagozzi, & Warshaw, 1989; Gefen, Straub, & Boudreau, 2000; Venkatesh, Morris, Davis, & Davis, 2003) to the context of CGM utilisation for travel planning. To measure source credibility factors, scales were adapted from Ohanian (1991). The *attitude* scale was adapted from Moon and Kim (2001).

Seven-point semantic differential scales were employed for the measurement of *perceived expertise*, *trustworthiness* and *attitude towards usage*, whereas the remaining constructs were measured using a seven-point Likert-type scale with anchors ranging from strongly disagree (1) to strongly agree (7). An expert panel review of scholars helped ensure content validity for the adapted scales. Panel members were provided with the definitions of the constructs and then asked to rate the appropriateness of the items in each scale. The “sum score” decision rule was employed due to its merit in taking into account all expert judges’ assessments.

3.2 Data Collection

Data were collected by means of a Web-based survey. Respondents were recruited through an online panel research company with a well-established panel management policy and an extensive filtering process. From a data quality perspective, it is desirable to take steps to ensure that only sample members could access the survey and that each respondent could complete the survey only once (Smyth & Pearson, 2011). To ensure this, sample members were assigned unique identifiers that have to be used to gain access to the survey. Employing the automatic login technique, each respondent's unique access code was embedded into the survey URL which was sent to them. This implies a unique URL link was generated for each e-mail address, and only one questionnaire was permitted from each respondent. *TripAdvisor* was chosen as the study's context because it is the largest and most popular travel-specific CGM site; hence respondents would have no problem understanding what the study was looking for. The online survey was opened for a period of 12 days. Within this period, 1,034 travel consumers from Singapore responded to the survey. From a data quality perspective, steps were taken to ensure that only sample members who have taken a leisure trip in the 12 months preceding the survey and who used the Internet for travel information search could proceed with the survey. In addition, each respondent could complete the survey only once.

3.2 Data Analysis

The resulting data were screened and prepared for analysis. Of a total of 1,034 respondents, 338 were initially screened out of the survey because they did not meet the sample criteria of (1) having taken leisure vacation within the previous 12 months and (2) having used the Internet to search for travel information. The data were further screened for incomplete

cases and outliers, leading to the deletion of 35 cases. After the screening process, a valid sample of 661 cases was retained for measurement analysis. For the estimation of the model, the partial least squares (PLS) path modelling technique was applied using the SmartPLS 2.0 (Beta) M3 software application (Ringle, Wende & Will, 2005). PLS was chosen over other alternative techniques due to its advantages – not requiring the distributional assumption of normality, making less demand on measurement scales, and working well with much smaller as well as much larger samples (Gefen *et al.*, 2000; Hair, Ringle, & Sarstedt, 2011).

4 Results

4.1 Respondents' Profile

The demographic characteristics of the respondents have been displayed in Table 1. Female respondents (53.1%) were slightly more than their male counterparts (46.9%). The respondents were fairly distributed across the different age groups with the clear majority aged between 20 and 60, representing the active working age group. About half of the respondents had a bachelor degree or a higher qualification. Though all the respondents had embarked on a leisure trip in the last 12 months preceding the survey, approximately 89% of them were certain of taking another trip in the subsequent year. The respondents were habitual Internet users with a vast majority using the Internet either several times each day (82.3%) or once daily (12.0%).

[Please Insert Table 1 here: Profile of study participants]

4.2 *Evaluation of the Measurement Model*

The adequacy of the measurement model was assessed by conducting PLS factorial validity tests. To validate the measures, convergent and discriminant validity were considered. The PLS factorial validity test offered evidence for convergent validity. The standardized factor loadings were all significant and greater than 0.70 (Table 2). The ρ and α of each latent construct exceeded 0.70 whereas the AVEs were all greater than 0.50. Also, discriminant validity of the measurement model was established. As shown in Table 3, the square root of AVE for each construct exceeded the correlations between the construct and other constructs (Fornell & Larcker, 1981). In addition, an examination of the indicators' cross loading revealed that no indicator loads higher on an opposing construct (Hair *et al.*, 2011). Thus, the measurement model parameter estimates and diagnostics provide strong evidence for the reliability and validity of construct measures.

[Please insert Table 2 here: Factor loadings for individual items]

[Please insert Table 3 here: Descriptive data, inter-construct correlations and the square root of AVE]

4.2 *Structural Model and Hypotheses Testing*

After establishing the validity of the measures, the structural model was assessed. Figure 1 displays the results of the structural model test. The findings indicate that the proposed model could explain up to 62.3 percent of the total variation in usage intention, 61.8 percent

of the variance in attitude and 41.9 percent of the total variation in usefulness perception.

The Stone–Geisser's Q^2 Test was also applied to ascertain the predictive validity of the exogenous latent variables. The resulting Q^2 values are significantly above zero (i.e. 0.517, 0.521 and 0.355 for usage intention, attitude and perceived usefulness, respectively), evident of the model's predictive relevance.

Following the procedure suggested by Hair et al. (2011), a bootstrap analysis was carried out. The non-parametric bootstrap analysis of 5,000 samples and 661 cases (using the individual sign change option) revealed that eight out of the eleven proposed relationships were significant. Table 4 displays the results of the hypotheses testing. *Hypothesis 1* which assume a direct positive relationship between perceived usefulness and attitude was supported ($\beta=0.120$, $t=2.607$, $p< 0.01$). Perceived usefulness was also found to have a direct positive effect on usage intention ($\beta=0.266$, $t=6.191$, $p< 0.01$). The perceived ease of using CGM platform was also found to be wielding a positive influence on attitude ($\gamma=0.416$, $t=10.808$, $p< 0.01$) and perception of usefulness ($\gamma=0.461$, $t=11.375$, $p< 0.01$). In addition, the present study found that trustworthiness positively drives perceived usefulness ($\gamma=0.248$, $t=5.346$, $p< 0.01$) and attitude ($\gamma=0.257$, $t=6.276$, $p< 0.01$) but its direct effect on usage intention was not supported ($\gamma=0.066$, $t=1.546$, $p>0.05$). Perceived expertise was also found to be directly related to attitude ($\gamma=0.166$, $t=4.677$, $p< 0.01$) but its relationships with perceived usefulness ($\gamma=0.041$, $t=1.107$, $p>0.05$) and usage intention ($\gamma=0.031$, $t=0.813$, $p>0.05$) were found to be insignificant. Lastly, *hypothesis 11* which posits a direct relationship between attitude and usage intention was also supported ($\beta=0.538$, $t=11.013$, $p< 0.01$).

[Please insert Fig. 2 here: Structural model test results]

5 Discussion and Conclusions

5.1 Key findings

The current study theorized that source credibility factors and technology acceptance factors would be predictors of online travellers' attitude and intention to use CGM for travel planning. To a large extent, the results of this empirical study lend support to the proposed model.

Several significant findings emerge. Most importantly, the TAM factors proved more salient in determining online travellers' acceptance of CGM for travel planning. The study finds that perceived usefulness has a direct influence as well as an indirect effect on usage intention through attitude towards usage. Online travellers' evaluation of the usefulness of CGM has its strongest direct impact on usage intention, an indication that travellers' develop their intentions towards the use of CGM for travel planning if they consider it as useful, over and above any positive or negative feelings they may have towards CGM usage for travel planning. This mirrors prior studies in other contexts which have also established a direct relationship between perceived usefulness and usage intention (Venkatesh *et al.*, 2003; Huh *et al.*, 2009; Casaló, Flavián, & Guinalíu, 2010)

The findings also highlight the critical role of ease of use perception in CGM contexts.

Perceived ease of use drives perceptions about the usefulness of CGM as well as the affective responses of online travellers towards CGM usage for travel planning. The results further show that perceived usefulness is most impacted by ease of use. This implies that online travellers' assessment of the amount of effort required when using CGM for travel planning considerably determines their attitudes as well as their perception of its usefulness.

Also, the findings of the present study demonstrate that source credibility factors are antecedents of CGM usefulness. Both perceived trustworthiness and expertise wield significant influence on usefulness perception. A recent study by Li (2013) recognises the influence of source credibility on cognitive and affective responses in the context of employees' acceptance of new information systems. In an another study on how knowledge workers are influenced to adopt the advice that they receive in mediated contexts, Sussman and Siegal (2003) also established that perceived credibility influences information usefulness. In addition, the present study finds that both trustworthiness and expertise significantly determine online travellers' attitudes toward the use of CGM for travel planning. This finding is consistent with the results of earlier studies on the relationship between source credibility and disposition toward information (Lafferty & Goldsmith, 1999; Kerstetter & Cho, 2004; Jin *et al.*, 2009). This would imply that online travellers are more favourably disposed towards the use of CGM for travel planning when they believe that CGM is from credible travellers. The study found that the strongest impact of perceived trustworthiness was on attitude towards using CGM for travel planning. Each

standard deviation variation in trustworthiness was associated with nearly 26 percent variation in attitude.

Nonetheless, the model has three insignificant relationships. First, perceived trustworthiness has no direct impact on usage intention in this context. Rather, the effect of trustworthiness on usage intention is mediated through attitude and usefulness. This result suggests that trustworthiness perception decides online travellers' behavioural intention to use CGM for travel planning by driving their perception of usefulness and negative/positive feelings towards this aspect of online behaviour. The mediating role of usefulness on the link between trustworthiness and intention was also confirmed by Sussman and Siegal's (2003) study on knowledge transfer via e-mails. Also, perceived expertise has direct effect on neither perceived usefulness nor usage intention. To some degree, this result differs from previous findings in certain contexts. Ohanian's (1991) study, for example, reported a significant direct influence of expertise on purchase intention, though the same study did not find the relationship between trustworthiness and purchase intention significant. This inconsistency could be due to the disparities in study contexts. For instance, whereas Ohanian's study regarded the source as celebrity endorsers, the present study refers to the source as travel consumers who post CGM on the *TripAdvisor* platform. Thus the findings may imply that, in the context of CGM, the dimensions of source credibility impact usage intention indirectly through their effect on perceived usefulness and attitude. Lastly, the study finds attitude in addition to usefulness as the direct predictors of usage intention. This mirrors previous research which highlighted the important role of attitude in deciding behavioural intention (Castañeda *et al.*, 2009; Morosan, 2012; Ayeh *et al.*, 2013).

5.2 *Implications and Limitations*

Undoubtedly, social media symbolize a dynamic online travel information source with enormous potential for influencing travel consumers (Ayeh, Leung, Au, & Law, 2012). This paper contributes to existing research by shedding light on the cognitive and affective processes underlying online travellers' acceptance of CGM for travel planning. A primary contribution is in combining technology acceptance and source credibility theories to examine travel consumers' attitudes and intention to use CGM for travel planning. By utilising technology acceptance literature to integrate and add to relevant concepts from the Source Credibility theory, this study offers a more in-depth understanding of online travellers' acceptance of CGM. In addition to verifying the theoretical validity of Davis' (1989) TAM to the context of CGM, the present study also provides a theoretical extension to explain online behaviour in this specific context. The findings suggest that source credibility factors effectively capture variations in perceived usefulness and attitudes that may not be accounted for by the TAM.

Not only do the source credibility factors explain additional variance, but also, they offer insights that can guide managerial action. By understanding online travellers' credibility perceptions in the present study's context, practitioners can design platforms that support effective CGM utilisation for travel planning. Thus, the proposed integrated model has practical value for guiding recommendations aimed at driving online travellers' acceptance of CGM for travel planning. First, practitioners should be aware of the critical effect of usefulness perception on travel consumers' affective and conative responses to the use of CGM for travel planning. Platform designers and managers can attempt to increase

perceptions about the usefulness of CGM by enhancing perceptions of ease of use and trustworthiness. To enhance ease of use perception, platform managers may need to focus on making CGM platforms user-friendly and augmenting usability. Platform managers should also aim to enhance trustworthiness by addressing credibility concerns. There is the need for CGM website managers to install effective mechanisms for ensuring a more secure reviewing platform where the best is done to guarantee credible CGM. Thus the insights this study offers can be translated into appropriate social media strategies for system designers.

While the TAM related constructs remain dominant factors in driving online travellers' attitude and intention to use CGM for travel planning, the present study provides support for the inclusion of source credibility factors to better understand the antecedents of perceived usefulness and attitude. Future studies can concentrate on further validation of these factors. Also, it is hoped that future studies might explore additional context-specific factors to better explain the complex relationships among variables in pursuit of understanding the use of CGM in travel planning contexts. The present study offers insights in this direction. Nevertheless, additional work is necessary to deepen the understanding of online behaviour regarding the dynamic nature of the varied CGM platforms in the travel domain. For example, it is possible for the effects of the credibility dimensions to vary across CGM platforms as a result of perhaps visual effects or other related factors.

It is worth noting that the cross-nature of this study and the reliance on self-report questionnaires to measure the proposed constructs imply that the results may be liable to common method variance. In addition, this study investigated only one CGM website (*TripAdvisor*), one that is widely used by travellers around the world. This may have

implications for the relative importance of cognitive factors such as trustworthiness and usefulness. Nevertheless, the results can be generalized to other CGM platforms which provide similar services. The extent to which the results could be generalized to non-travel-specific CGM platforms however remains to be found out.

Overall, this study provides an appropriate theoretical framework for studying the complex phenomenon of CGM use for travel planning and offers insights to practitioners on social media marketing strategies within the travel domain. In concert with the empirical findings, the proposed model can serve as a valuable foundation for future research regarding this vital aspect of online behaviour.

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FIGURES

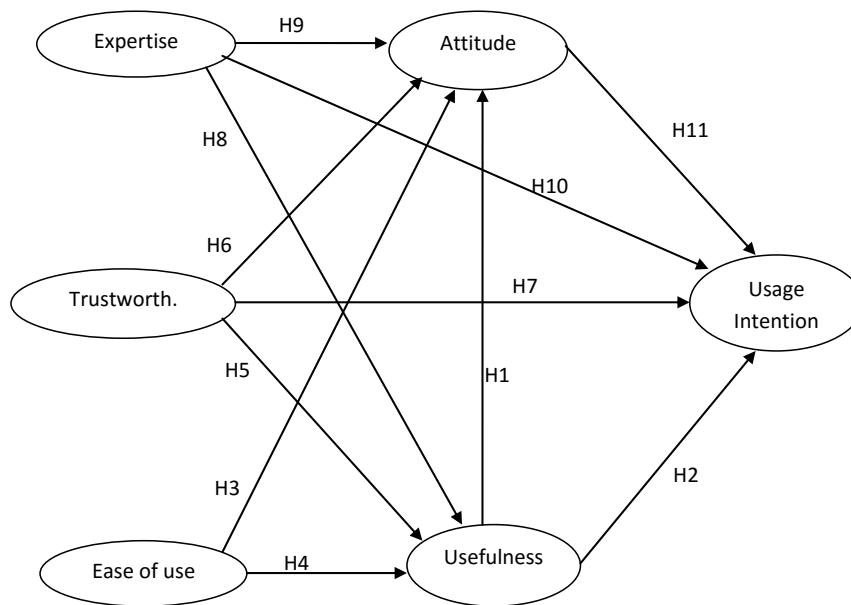
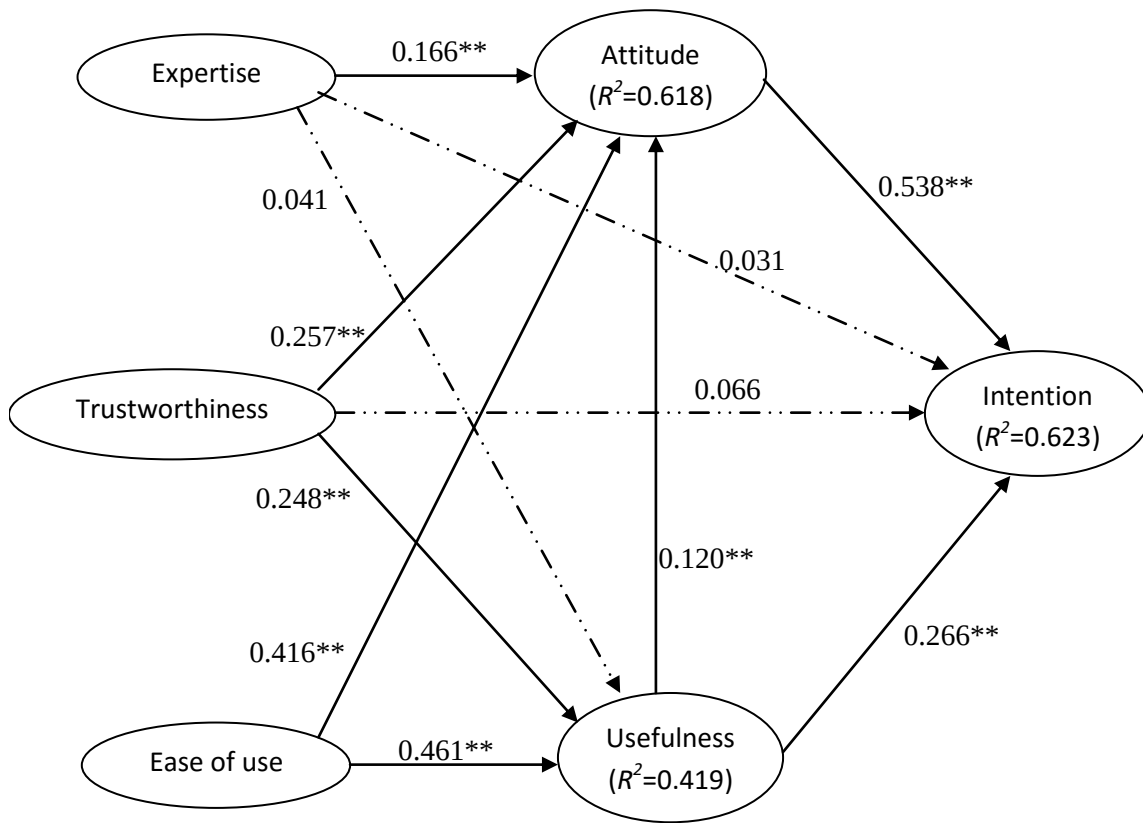


Fig. 1. Proposed model with hypothesised relationships



**Significant at $p < 0.01$

Fig. 2. Structural model test results

TABLES

Table 1. Profile of study participants (N=661)

<i>Profile category</i>		<i>Frequency</i>	<i>Percentage (%)</i>
Gender	Female	351	53.1
	Male	310	46.9
Age	Less than 20 years	26	3.9
	21-30 years	168	25.4
	31-40 years	215	32.5
	41-50 years	162	24.5
	51-60 years	76	11.5
	60+ years	14	2.1
Education	High school/ Secondary school	132	20.0
	Diploma/Associate degree (2 years)	185	28.0
	Bachelor degree	196	29.7
	Graduate/Postgraduate degree	138	20.9
	Others	10	1.5
Embarking on a vacation trip in the next 12 months	Yes	588	89.2
	Not Sure	63	9.6
	No	8	1.2
Internet use frequency	A few times in a month or less	6	0.9
	A few times in a week	32	4.8
	About once a day	79	12.0
	Several times on each day	544	82.3

Table 2. Factor loadings for individual items

<i>Construct/Item</i>	<i>Loading</i>	<i>Cronbach's Alpha (α)</i>	<i>Composite Reliability</i>	<i>AVE</i>
<i>Attitude:</i>		0.956	0.966	0.851
Bad / Good	0.915			
Foolish / Wise	0.930			
Unpleasant / Pleasant	0.931			
Negative / Positive	0.914			
Unfavourable/ Favourable	0.923			
<i>Ease of use:</i>		0.953	0.964	0.843
Easy to learn how to use	0.904			
Easy to use to find the information needed	0.924			
Easy for me to become skilful at using	0.901			
Easy to use content to plan my trips	0.918			
Overall, I find easy to use	0.942			
<i>Expertise:</i>		0.93	0.947	0.782
Inexperienced / Experienced in travel	0.868			
Not experts / Experts in travel	0.881			
Unknowledgeable / Knowledgeable in travel	0.905			
Unqualified / Qualified to offer travel advice	0.858			
Unskilled / Skilled in travel	0.908			
<i>Intention:</i>		0.953	0.964	0.841
Will not hesitate to visit	0.882			
Plan to seek travel advice from	0.909			
Expect to use to plan my future trips	0.937			
Intend to use to inform my travel-related decisions	0.920			
Very likely to use for my travel planning	0.937			
<i>Trustworthiness:</i>		0.934	0.95	0.792
Undependable / Dependable	0.819			
Dishonest / Honest	0.908			
Unreliable / Reliable	0.916			
Insincere / Sincere	0.900			
Untrustworthy / Trustworthy	0.903			
<i>Usefulness:</i>		0.956	0.966	0.851
Improves my travel planning	0.921			
Helps me plan trips more efficiently	0.933			
Makes my travel planning easier	0.917			
Makes it easier for me to reach travel-related decisions	0.909			
Overall, I find useful for travel planning	0.932			

Note: All loadings are significant at $p < 0.001$.

Table 3. Descriptive data, Inter-construct correlations and the square root of AVE

<i>Constructs:</i>	<i>Attitude</i>	<i>Ease of use</i>	<i>Expertise</i>	<i>Intention</i>	<i>Trustworthiness</i>	<i>Usefulness</i>
Attitude	(0.922)					
Ease of use	0.689	(0.918)				
Expertise	0.589	0.446	(0.884)			
Intention	0.751	0.739	0.51	(0.917)		
Trustworthiness	0.643	0.492	0.722	0.569	(0.890)	
Usefulness	0.571	0.601	0.426	0.62	0.504	(0.922)
<i>Mean</i>	<i>5.507</i>	<i>5.543</i>	<i>4.788</i>	<i>5.500</i>	<i>5.068</i>	<i>5.458</i>
<i>Standard Deviation</i>	<i>1.011</i>	<i>0.997</i>	<i>0.997</i>	<i>1.036</i>	<i>0.983</i>	<i>1.048</i>

Note: The number in parenthesis is the square root of AVE.

Table 4. Results of hypothesis testing

<i>Hypotheses</i>	<i>Path coefficient</i>	<i>t-value</i>	<i>Supported?</i>
H1: Usefulness → Attitude	0.120**	2.607	Yes
H2: Usefulness → Intention	0.266**	6.191	Yes
H3: Ease of use → Attitude	0.416**	10.808	Yes
H4: Ease of use → Usefulness	0.461**	11.375	Yes
H5: Trustworthiness → Usefulness	0.248**	5.346	Yes
H6: Trustworthiness → Attitude	0.257**	6.276	Yes
H7: Trustworthiness → Intention	0.066	1.546	No
H8: Expertise → Usefulness	0.041	1.107	No
H9: Expertise → Attitude	0.166**	4.677	Yes
H10: Expertise → Intention	0.031	0.813	No
H11: Attitude → Intention	0.538**	11.013	Yes

**Significant at $p < 0.01$.

Note: These significance levels are determined via bootstrapping analysis (Hair *et al.*, 2011)